Self Similar Model analysis

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1 Introduction

The general surface density profiles is

$$\sigma(x) = \frac{\sigma_c}{\sqrt{1 + (r/R)^n}}$$

However, in our self-similar model, we observed that asymptotically
This equation has asymptotic ($r \gg R$) form

$$\sigma(r) = \sigma_0 \left(\frac{r}{R}\right)^{-1.4} \propto r^{-1.4}.$$

We know that the Mass of the disk has the relation

$$M(r) = \int 2\pi r' \sigma(r') dr'$$

So, for our current surface density profile we get the mass of the disk as

$$M(r) = \int 2\pi r' \sigma_0 \left(\frac{r'}{R}\right)^{-1.4} dr'$$

$$M(r) = \frac{2\pi\sigma_0}{R^{-1.4}} \int r'^{-0.4} dr'$$

$$M(r) = \frac{2\pi\sigma_0}{0.6\, R^{-1.4}} (r^{0.6})$$

after simplifying we get,

$$M(r) = \frac{10\pi\sigma_0 R^{1.4}}{3} r^{0.6}$$